

# Scope of Work: Pose Estimation Internship Project (Ball Tracking Initiative)

## Objective

To enhance the ball tracking system by integrating accurate pose estimation models that can detect and reconstruct human body posture and limb positions during cricket matches. This module will aid in refining decision-making scenarios like LBWs, catches, and close-in fielding actions.

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## Project Components

### 1. Data Acquisition & Preprocessing

- **Footage Preparation:**
    - Annotated footage from previous tests and stadium deployments (e.g., BITS Pilani).
    - Multi-angle synchronized videos to allow multi-view pose triangulation.
  - **Data Formatting:**
    - Standardize frame formats (preferably JPEG 2000) to ensure compatibility with DALI inference pipelines.
    - Include metadata tagging (camera ID, timestamps, event type).
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### 2. Model Architecture & Training

- **Pose Estimation Frameworks:**
  - Start with pretrained models (e.g., OpenPose, HRNet, MediaPipe).
  - Fine-tune using cricket-specific datasets to improve robustness under occlusions (e.g., gloves, helmets, shadows).

- **Multi-View Fusion:**
    - Use multi-camera calibration data to reconstruct 3D joint positions.
    - Integrate triangulation logic similar to the one used in ball trajectory estimation.
  - **Temporal Smoothing:**
    - Implement temporal consistency using Kalman filters or LSTM-based post-processing to smooth out frame-to-frame jitter.
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### 3. System Integration

- **Calibration Alignment:**
    - Align pose keypoints with intrinsic and extrinsic calibration models being used in the ball tracking system.
    - Validate against Total Station measurements where applicable.
  - **Overlay & Visualization:**
    - Develop basic overlays showing skeletal joints on live footage.
    - Enable side-by-side comparison between human-detected pose and expected biomechanics.
  - **Modular Deployment:**
    - Design the pipeline as a plug-and-play module for Unreal Engine integration or HyperView use.
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### 4. Performance Metrics & Validation

- **Quantitative Benchmarks:**
  - 2D Reprojection Error: Target < 10 pixels per joint.

- 3D Keypoint Error: < 25mm deviation from ground-truth markers.
  - **Use-case Specific Accuracy:**
    - Bat-swing dynamics
    - Keeper reaction time
    - Fielder position estimation in close calls
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## Tools & Technologies

- **Frameworks:** PyTorch, TensorFlow, OpenCV, DALI
  - **Visualization:** Unreal Engine (UE5), HyperView 1.0
  - **Storage:** NVMe SSDs, JPEG 2000 image format
  - **Calibration:** OpenCV calibration toolkits, Bundle Adjustment
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## Research & Experimentation Opportunities

- Biomechanical analysis from pose data (e.g., bowling action comparison)
  - Integration of pose data for AI-generated storytelling and highlight reels.
  - Real-time operator overlay suggestions for DRS scenarios.
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## Team Involvement & Reporting

- **Primary Mentors:** Aafaq Inamdar (Dev), Vedanshu (Triangulation), Siddhant (UI/UX)
- **Weekly Check-ins:** Interns to demo progress and receive feedback

- **Monthly Evaluation:** Accuracy benchmarking and system performance audits

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## **Success Metrics**

- **Real-time estimation (< 200ms latency per frame).**
- **Sub-25mm average 3D keypoint error.**
- **Integration with 2+ camera viewpoints and full DRS pipeline readiness.**